

## HOTING, Sweden

WMO: 022420

Lat: 64.088N

Lon: 16.236E

Elev: 241

StdP: 98.46

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB   |              | Humidification DP/MCDB and HR |            |              |              |            |              | Coldest Month WS/MCDB |            |            |             | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|--------------|--------------|-------------------------------|------------|--------------|--------------|------------|--------------|-----------------------|------------|------------|-------------|-----------------------|------------|--------------|
|               |              |              | 99.6%                         |            |              | 99%          |            |              | 0.4%                  |            | 1%         |             |                       |            |              |
|               | 99.6%        | 99%          | DP                            | HR         | MCDB         | DP           | HR         | MCDB         | WS                    | MCDB       | WS         | MCDB        | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-29.3 | (c)<br>-26.1 | (d)<br>-32.5                  | (e)<br>0.2 | (f)<br>-29.2 | (g)<br>-29.0 | (h)<br>0.3 | (i)<br>-26.1 | (j)<br>6.9            | (k)<br>1.2 | (l)<br>5.9 | (m)<br>-0.9 | (n)<br>0.2            | (o)<br>110 | (p)<br>0.636 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month<br>DB Range | Cooling DB/MCWB |             |             |             |             |             | Evaporation WB/MCDB |             |             |             |             |             | MCWS/PCWD<br>to 0.4% DB |            |
|---------------|---------------------------|-----------------|-------------|-------------|-------------|-------------|-------------|---------------------|-------------|-------------|-------------|-------------|-------------|-------------------------|------------|
|               |                           | 0.4%            |             | 1%          |             | 2%          |             | 0.4%                |             | 1%          |             | 2%          |             |                         |            |
|               |                           | DB              | MCWB        | DB          | MCWB        | DB          | MCWB        | WB                  | MCDB        | WB          | MCDB        | WB          | MCDB        | MCWS                    | PCWD       |
| (a)<br>7      | (b)<br>9.7                | (c)<br>25.7     | (d)<br>16.6 | (e)<br>23.6 | (f)<br>15.9 | (g)<br>21.5 | (h)<br>15.0 | (i)<br>18.1         | (j)<br>23.1 | (k)<br>17.0 | (l)<br>21.7 | (m)<br>16.0 | (n)<br>20.2 | (o)<br>2.1              | (p)<br>150 |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.3                            | 11.9 | 19.8 | 15.2 | 11.1 | 18.8 | 14.1 | 10.3 | 17.9 | 52.2          | 23.1 | 48.8 | 21.7 | 45.4 | 20.2 | 21.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 6.1           | (b) 5.3 | (c) 4.5 | (e) -32.1                  | (f) 27.7 | (g) 3.0            | (h) 2.0 | (i) -34.3                                          | (j) 29.1 | (k) -36.1  | (l) 30.3 | (m) -37.8  | (n) 31.4 | (o) -40.0  | (p) 32.8 |    |    |
| (a) 6.1           | (b) 5.3 | (c) 4.5 | (e) -32.2                  | (f) 19.5 | (g) 2.8            | (h) 1.4 | (i) -34.2                                          | (j) 20.5 | (k) -35.9  | (l) 21.3 | (m) -37.4  | (n) 22.1 | (o) -39.5  | (p) 23.2 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 2.4    | -9.9  | -8.9  | -4.5  | 1.8   | 7.7   | 12.1  | 15.5  | 13.5  | 8.7   | 2.5   | -2.9  | -7.2 |
|                                                                                  | DBStd       | 10.02  | 7.73  | 7.31  | 5.89  | 3.43  | 3.67  | 3.28  | 3.12  | 2.85  | 2.98  | 3.76  | 5.08  | 7.27 |
|                                                                                  | HDD10.0     | 3161   | 617   | 528   | 451   | 245   | 91    | 13    | 1     | 4     | 59    | 233   | 387   | 533  |
|                                                                                  | HDD18.3     | 5817   | 876   | 761   | 709   | 495   | 329   | 187   | 99    | 153   | 289   | 491   | 637   | 791  |
|                                                                                  | CDD10.0     | 400    | 0     | 0     | 0     | 0     | 21    | 78    | 171   | 111   | 19    | 1     | 0     | 0    |
|                                                                                  | CDD18.3     | 15     | 0     | 0     | 0     | 0     | 0     | 2     | 11    | 2     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 200    | 0     | 0     | 0     | 0     | 8     | 46    | 122   | 24    | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 24     | 0     | 0     | 0     | 0     | 0     | 3     | 19    | 1     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 1.9    | 1.5   | 1.6   | 2.0   | 1.9   | 2.3   | 2.4   | 2.0   | 1.8   | 1.9   | 1.8   | 1.6   | 1.6  |
| Precipitation                                                                    | PrecAvg     | 629    | 48    | 29    | 28    | 34    | 42    | 68    | 94    | 84    | 56    | 53    | 50    | 43   |
|                                                                                  | PrecMax     | 818    | 97    | 56    | 54    | 77    | 94    | 140   | 187   | 172   | 145   | 148   | 112   | 82   |
|                                                                                  | PrecMin     | 461    | 9     | 4     | 7     | 10    | 13    | 14    | 22    | 26    | 18    | 21    | 21    | 8    |
|                                                                                  | PrecStd     | 113    | 21    | 14    | 11    | 16    | 23    | 30    | 44    | 46    | 27    | 28    | 26    | 17   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 5.5   | 6.4   | 8.0   | 17.4  | 24.4  | 27.1  | 29.0  | 25.9  | 19.2  | 13.7  | 9.2   | 5.7  |
|                                                                                  |             | MCWB   | 3.6   | 4.0   | 4.4   | 9.1   | 14.1  | 15.5  | 18.3  | 17.4  | 14.3  | 10.4  | 7.9   | 3.8  |
|                                                                                  | 2%          | DB     | 3.7   | 4.8   | 6.2   | 13.0  | 21.2  | 24.7  | 26.5  | 23.6  | 16.8  | 10.5  | 6.1   | 4.1  |
|                                                                                  |             | MCWB   | 2.3   | 2.8   | 3.2   | 6.8   | 12.5  | 15.0  | 17.3  | 16.7  | 13.1  | 8.9   | 5.2   | 2.7  |
|                                                                                  | 5%          | DB     | 1.9   | 2.8   | 4.8   | 10.3  | 18.5  | 21.5  | 24.4  | 21.3  | 15.0  | 9.3   | 4.4   | 2.7  |
|                                                                                  |             | MCWB   | 0.9   | 1.3   | 2.4   | 5.4   | 11.1  | 13.7  | 16.7  | 15.9  | 12.0  | 8.0   | 3.5   | 1.7  |
|                                                                                  | 10%         | DB     | 0.2   | 0.8   | 3.5   | 8.1   | 15.4  | 19.2  | 22.2  | 19.2  | 13.6  | 8.1   | 3.1   | 1.4  |
|                                                                                  |             | MCWB   | -0.4  | -0.2  | 1.4   | 4.2   | 9.7   | 12.7  | 15.7  | 14.9  | 11.2  | 6.8   | 2.3   | 0.7  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 3.7   | 4.2   | 4.9   | 9.5   | 15.0  | 17.6  | 19.9  | 19.0  | 15.1  | 10.8  | 8.1   | 4.1  |
|                                                                                  |             | MCDB   | 5.0   | 6.1   | 7.4   | 16.1  | 21.9  | 23.5  | 24.6  | 23.4  | 17.9  | 12.6  | 9.4   | 5.4  |
|                                                                                  | 2%          | WB     | 2.4   | 2.9   | 3.6   | 7.5   | 13.5  | 15.9  | 18.6  | 17.6  | 13.6  | 9.3   | 5.4   | 2.8  |
|                                                                                  |             | MCDB   | 3.6   | 4.5   | 5.7   | 12.2  | 19.5  | 22.0  | 23.9  | 21.7  | 15.9  | 10.2  | 6.2   | 3.8  |
|                                                                                  | 5%          | WB     | 1.0   | 1.5   | 2.6   | 6.0   | 11.9  | 14.6  | 17.7  | 16.5  | 12.6  | 8.2   | 3.7   | 1.8  |
|                                                                                  |             | MCDB   | 1.7   | 2.7   | 4.5   | 9.4   | 17.6  | 20.3  | 22.8  | 20.2  | 14.4  | 9.3   | 4.3   | 2.7  |
|                                                                                  | 10%         | WB     | -0.3  | -0.2  | 1.5   | 4.5   | 10.2  | 13.4  | 16.6  | 15.3  | 11.7  | 6.9   | 2.4   | 0.7  |
|                                                                                  |             | MCDB   | 0.3   | 0.7   | 3.3   | 7.6   | 14.8  | 18.1  | 21.3  | 18.7  | 13.2  | 7.9   | 3.0   | 1.3  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 8.1   | 9.1   | 10.6  | 9.7   | 10.5  | 9.9   | 9.7   | 9.1   | 7.7   | 5.7   | 4.9   | 7.0  |
|                                                                                  |             | MCDBR  | 8.0   | 7.4   | 9.2   | 14.7  | 16.4  | 14.3  | 13.6  | 13.0  | 10.0  | 7.5   | 5.2   | 7.1  |
|                                                                                  |             | MCWBR  | 7.2   | 6.3   | 6.7   | 8.8   | 8.4   | 6.5   | 5.9   | 6.8   | 6.6   | 5.7   | 4.6   | 6.4  |
|                                                                                  | 5% WB       | MCDBR  | 7.9   | 7.1   | 8.9   | 12.9  | 14.8  | 12.7  | 11.6  | 11.1  | 8.8   | 7.1   | 5.0   | 6.9  |
|                                                                                  |             | MCWBR  | 7.2   | 6.2   | 6.7   | 8.1   | 8.3   | 6.3   | 5.7   | 6.2   | 6.1   | 5.7   | 4.6   | 6.2  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.202  | 0.270 | 0.298 | 0.334 | 0.338 | 0.336 | 0.349 | 0.338 | 0.310 | 0.293 | 0.348 | 0.352 |      |
|                                                                                  | taud        | 1.996  | 2.204 | 2.276 | 2.363 | 2.466 | 2.498 | 2.476 | 2.497 | 2.563 | 2.450 | 2.743 | 2.731 |      |
|                                                                                  | Ebn at Noon | 380    | 650   | 793   | 844   | 875   | 884   | 859   | 837   | 795   | 639   | 366   | 203   |      |
|                                                                                  | Edh at Noon | 26     | 57    | 82    | 95    | 94    | 93    | 93    | 83    | 63    | 46    | 23    | 15    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.15   | 0.73  | 2.12  | 3.81  | 4.71  | 5.29  | 4.84  | 3.64  | 2.12  | 0.89  | 0.22  | 0.06  |      |
|                                                                                  | RadStd      | 0.01   | 0.11  | 0.23  | 0.31  | 0.51  | 0.40  | 0.54  | 0.35  | 0.25  | 0.14  | 0.04  | 0.00  |      |

## Historical Trends

|                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only            | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (45 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                         | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | -137        | N/A     | N/A     | N/A     |

Nomenclature: See separate page